

The Impact of Interdisciplinary, Gender and Geographic Distributions on the Citation Patterns of the Journal of Chemical Information and Modeling

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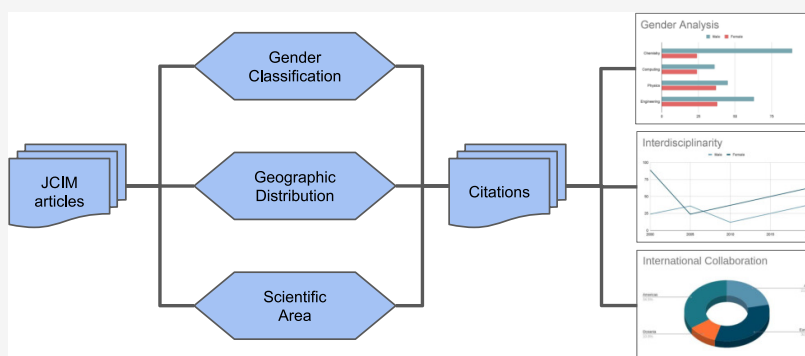


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ABSTRACT: There has been a growing recognition of the need for diversity and inclusion in scientific fields. This trend is reflected in the Journal of Chemical Information and Modeling (JCIM), where there has been a gradual increase in the number of papers that embrace this diversity. In this viewpoint, we analyze the evolution of the profile of papers published in JCIM from 1996 to 2022 addressing three diversity criteria, namely interdisciplinarity, geographic and gender distributions, and their impact on citation patterns. We used natural language processing tools for the classification of main areas and gender, as well as metadata, to analyze a total of 7384 articles published in the categories of research articles, reviews, and brief reports. Our analyses reveal that the relative number of articles and citation patterns are similar across the main areas within the scope of JCIM, and international collaboration and publications encompassing two to three research areas attract more citations. The percentage of female authors has increased from 1996 (less than 20%) to 2022 (more than 32%), indicating a positive trend toward gender diversity in almost all geographic regions, although the percentage of publications by single female authors remains lower than 20%. Most JCIM citations come from Europe and the Americas, with a tendency for JCIM papers to cite articles from the same continent. Furthermore, there is a correlation between the gender of the authors, as JCIM manuscripts authored by females are more likely to be cited by other JCIM manuscripts authored by females.

1. INTRODUCTION

The primary objective of scientific publications is to advance knowledge by distributing research findings in a transparent, rigorous, and accessible manner to the scientific community. Typically, academic journals employ a peer review process that plays a crucial role in standards of rigor, ethics, and transparency in scientific publications. In addition, academic journals should also reach and resonate with a wide range of authors and readers in the scientific community. However, unconscious bias and lack of diversity in editorial boards or reviewer pools can flaw peer review and affect the reliability and credibility of scientific publications. The promotion of greater diversity of editorial boards and reviewer pools is expected to lead to more inclusive and balanced decision-

making processes, therefore reducing biases in publication selection and contributing to a more equitable and representative scientific community, particularly considering the natural disparity in gender and resources across different regions. Although these actions can be applied by different academic journals, a quantitative assessment of their success is

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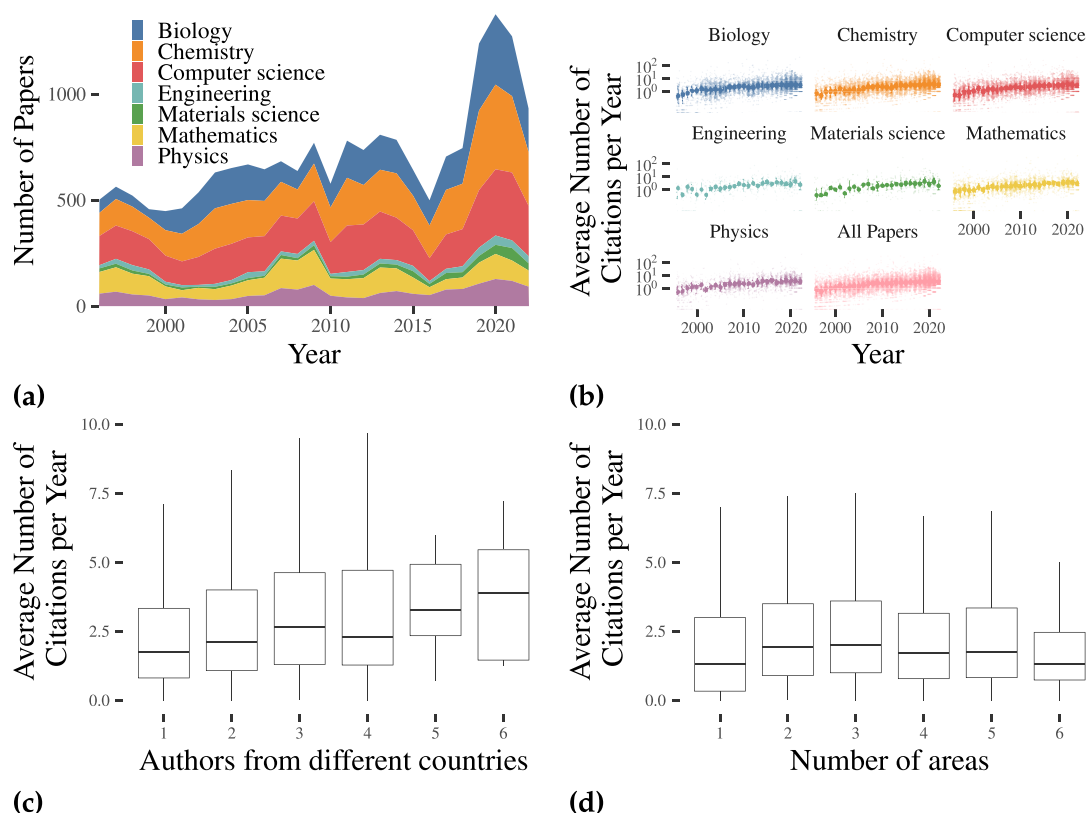


Figure 1. Diversity of the main research fields and internationalization of JCIM publications. Time evolution of (a) general research fields published in JCIM, and the normalized (by years from publication) number of citations (b) within a given subject field, (c) per number of different main subjects and (d) per number of authors of different countries.

a challenge, although some quantitative studies have recently been conducted, mainly with a focus on gender inequality.^{1–4}

We contribute to the discussion by examining JCIM publications with respect to three diversity criteria, namely interdisciplinarity, geographic and gender distributions, and their impact on citation patterns. We considered as a data set the papers published between January 1996 (volume 36, number 1) and October 2022 (volume 62, number 19). From the total of 7622 published papers, we excluded editorials, book reviews, corrections, replies, rapid communications, article commentaries, and retractions from the data set (according to the classification of ACS metadata), resulting in a total of 7384 analyzed.

We obtain the categories of associated concepts for each paper from the OpenAlex⁵ database. OpenAlex uses data from CrossRef⁶ and enriches these data with additional information. Some of this information corresponds to “concepts” appearing in the document, according to the Microsoft Academic Graph (MAG) ontology.⁷ This ontology corresponds to a knowledge graph of concepts, which refers to an abstract or general idea that represents a class of entities or things within a semantically linked domain of knowledge. At the top level, the MAG ontology corresponds to the main research areas. The Microsoft Academic Graph ontology has 19 root-level areas and up to six layers of descendants.

The concept tagging is based on a Natural Language Processing model that takes into account the title, abstract, type, and journal of the article, and uses an attention-based neural network to attribute the concepts using a multilabel classifier. If a concept is predicted to be present in a paper, the corresponding root-level area is assigned to the paper. In this

work, we examine the root level of seven main concepts that correspond to the main general fields published in JCIM (biology, chemistry, computer science, engineering, materials science, mathematics, and physics). The model has high accuracy for biology, chemistry and materials science (above 70%) and considerable accuracy for computer science, engineering, mathematics and physics (between 60% and 70%).

OpenAlex also provides the last known affiliation of each author. The authors are disambiguated using information such as ORCID, affiliation, and coauthorship. For the last known affiliations, we retrieved country information to measure levels of international collaboration.

We use a name-based method to predict the gender of the authors. The model is fine-tuned from Bidirectional Encoder Representations for Transformers (BERT) using the data from.⁸ The data are made up of author names and genders taken from different sources, such as zbMATH, Web of Science, and Pubmed. The model achieved an overall accuracy of 84.2% and a precision greater than 80% for both genders, a value comparable to the state-of-the-art methods. A known drawback of these models is that the cultural context of a name affects gender inference, and Asian names had less certain predictions.⁸

2. RESULTS AND DISCUSSION

We begin by analyzing the diversity of research areas and the internationalization of JCIM publications in Figure 1.

We observe that the ratio of publications in the main research areas covered by JCIM has been stable over the years, with an increase in the number of articles in the fields of

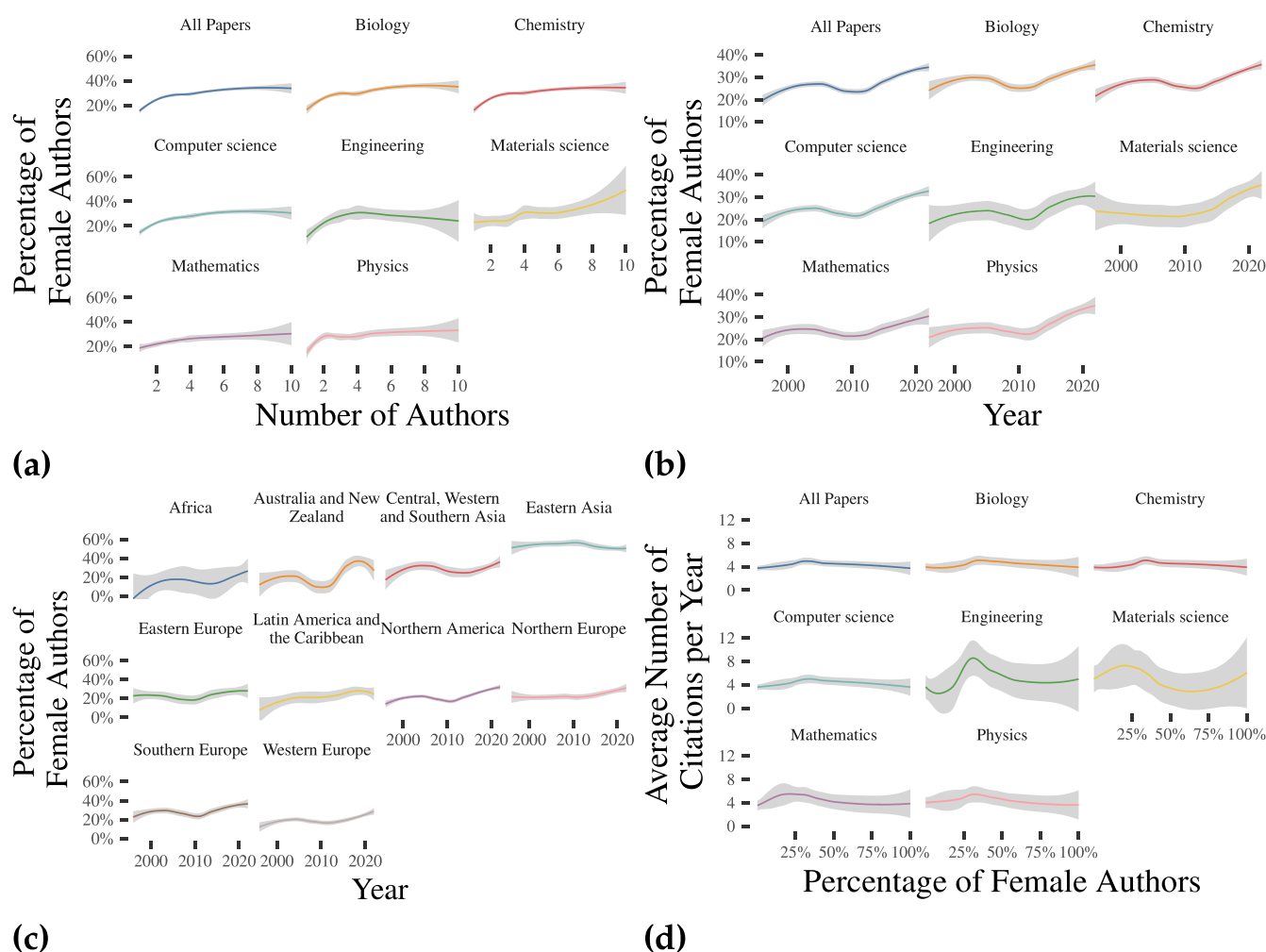


Figure 2. Female authorship distributions and citations. (a) The percentage of female authors according to the number of authors, (b) the evolution of the ratio of female authors over years, (c) the evolution of the percentage of female authors per geographic region, (d) the percentage of female authors versus the average number of citations (normalized by the years of publication of the article).

biology, chemistry and computer science throughout the past decade (Figure 1a). This trend follows an increase in the number of articles published in the same period. There is also a steady increase in the average annual citation among the different main areas (Figure 1b) and in the average number of citations for articles affiliated with different countries (Figure 1c). This is shown by the increase in the median (central line in the box plot) of the number of average citations with the number of authors affiliated with institutions from different countries (Figure 1c). Interdisciplinary studies with up to three main research areas attract more citations, although articles covering a larger number of areas (>4) receive fewer citations (Figure 1d). Papers that encompass six different areas have a similar distribution of citations as papers that focus on a single area. In summary, we observe that the relative number of articles and citation pattern are similar across areas and international collaboration and publications covering two to three main areas attract more citations.

Next, we analyze the gender and geographic distributions of JCIM authors (Figure 2). The percentage of female authors ranges between 30% and 40% for manuscripts with two or more authors, while the percentage of articles with a single female author is around 20% (Figure 2a). This pattern is consistent for all main fields (Figure 2a). There is also a

significant increase in the percentage of female authors in recent years (Figure 2b) with a peak in some fields between 2000 and 2005. This may have been due to the publication of special issues related to conferences in the period prior to the change in the name of JCIM from Journal of Chemical Information and Computer Sciences. The increase in the percentage of female authors is observed not only for all main fields (Figure 2b) but for all geographic regions (Figure 2c). Yet, the unusually high percentages of female authors of articles from Eastern Asia is likely due to ambiguities associated with the fact that in many Asian languages, names are often nongender specific.⁹

Annual citation of JCIM articles according to the percentage of female authors indicates that mixed gender coauthorship receives slightly more citations (Figure 2d). Papers authored predominantly by male or female authors tend to receive fewer citations, particularly in the fields of biology, chemistry, and computer science. Annual citation of JCIM articles according to the percentage of female authors indicates lower citation rates than articles authored by male counterparts (Figure 2d). In fact, as the percentage of female authors increases, the number of citations decreases. The same trend was previously reported from the assessment of the publication stages in chemical science journals published by the RSC journals from

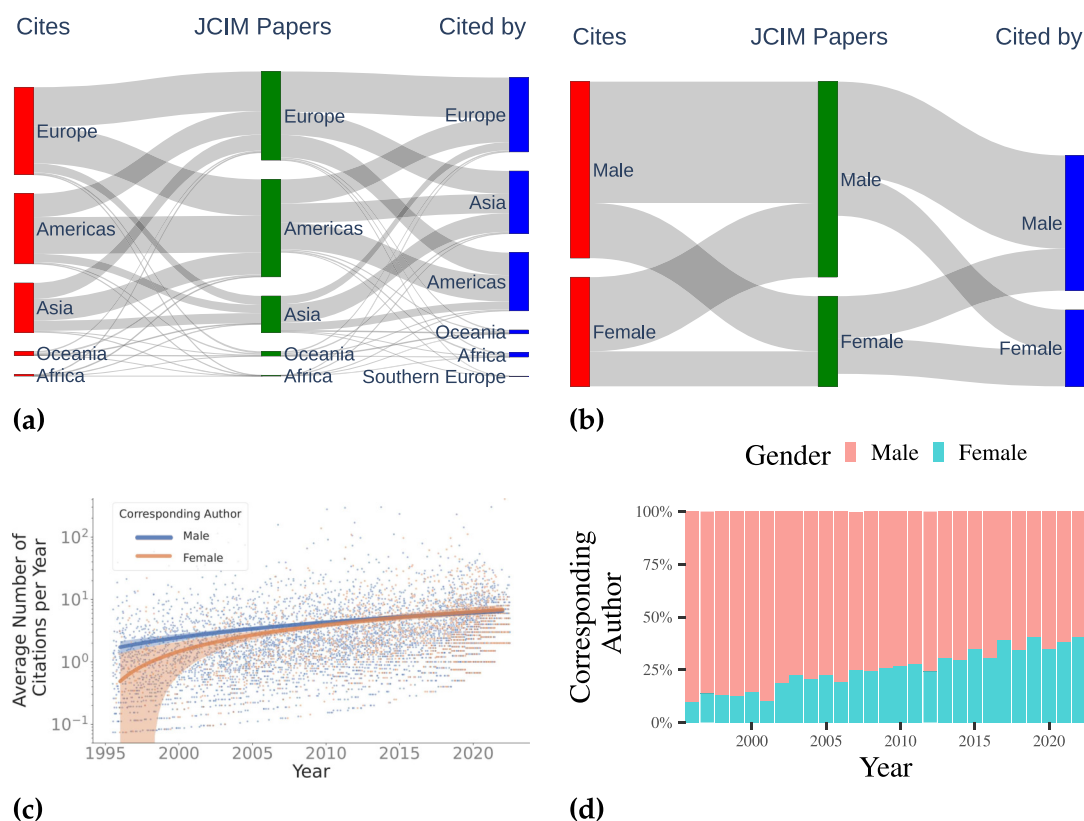


Figure 3. Citation patterns by geographical location and gender and the corresponding author by gender. Distribution of the number of papers that cite (red) or were cited (blue) by JCIM manuscripts (green) according to (a) geographical affiliation and (b) gender. Normalized (by years from publication) number of citations according to the gender of the corresponding author (c) and evolution of gender proportion of the corresponding author over the years (d).

2014 to 2018,⁹ but in contrast to an Elsevier report that found that while female researchers publish fewer articles than their male counterparts, there was no difference in the number of citations between genders.¹⁰

Lastly, we analyze the citation patterns for papers as a function of the geographic and gender distribution of who cited whom and the corresponding authorship gender (Figure 3). The flow of citations of articles cited by (in blue) and that cites (in red) articles published in JCIM (in green) from geographic regions (Figure 3a) and gender (Figure 3b). The flow of citations exhibited two clear trends: a predominance of citations from Europe and the Americas, and a larger proportion of papers being cited or citing articles from the same continent.

Regarding the gender distribution of the JCIM citations, articles with female authorship are proportionally more cited by articles that also have female authorship (Figure 3b). This homophilic characteristic is observed in different areas,⁴ and is also reported in RSC publications.⁹ This suggests that an increase in the number of female authors' publications may lead to greater gender equity in paper citations. In any case, the number of JCIM papers with corresponding female authors has been continuously increasing (Figure 3d) and has received an increasing number of citations, particularly in the past decade (Figure 3c). In fact, it can be seen in Figure 3d that the most cited paper of 2022 has a female researcher as the corresponding author. This trend is not in line with the global trend,² although it is reported that the gender citation imbalances vary between journals.⁴ These trends may have benefited from special issues of JCIM dedicated to female

contributions to computational chemistry,^{11,12} which showcased the excellence and diversity of female research in the field.

3. CONCLUSIONS

We have evaluated the impact of selected diversity criteria on the citations of papers published in JCIM from 1996 to 2022. We conclude that, in general, greater diversity of geographic distribution and interdisciplinarity promoted increased citations. For example, publications from authors affiliated with different countries received more citations than those from a single country affiliation. Interdisciplinary papers in two to three research areas resulted in slightly more citations compared to papers belonging to one or more than three research areas. We also observed a continuous increase in the percentage of female authors over time and in the number of citations received by papers with female corresponding authors. Finally, papers authored by females are more cited by other female authors than by their male counterparts, though females authors cite slightly more male than female authors. Regarding future work, it would be interesting to compare these results with those of other journals.

■ ASSOCIATED CONTENT

Data Availability Statement

Full-length articles from JCIM were made available through an agreement established between ACS and the authors. OpenAlex metadata was retired from their API using the openalexR package available on the CRAN Web site.¹³ The concept tagging model is publicly available on GitHub.¹⁴ The gender

classification model available in Huggingface.¹⁵ All data and scripts to reproduce the figures in this paper are available on figshare.¹⁶

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Notes

The authors declare no competing financial interest.

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